

Program : Diploma in Computer Application and Business Management	
Course Code : 3251	Course Title: Computer Application in Management
Semester : 3	Credits: 3
Course Category: Foundation Course	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To provide the students with the theoretical knowledge to deal with the basics of computer hardware and software.
- To lay foundation in basic aspects like operating systems, data processing, data communication, computer networks etc which is very well required in the management of an organization.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basics of Computer		Office Automation Software Lab	1

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of computer systems starting from the evolution, basic architecture and components	12	Understanding
CO2	Identify the different Operating Systems and their functions	10	Applying
CO3	Utilize the concepts of Data Processing and Data Communication	11	Applying
CO4	Interpret the basics of Computer Networks and Internet	10	Applying
	Series Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2	3						
CO3	3						
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of computer systems starting from the evolution, basic architecture and components.		
M1.01	Know the evolution, characteristics, classification, and generations of computers.	3	Remembering
M1.02	Understand the basic architecture of a computer, the different components of computer, computer memory and types of memory.	4	Understanding
M1.03	Identify the types of data representations in computers, hardware, software, and types of software, and differentiate between system software and application Software, Free and open source software	2	Understanding
M1.04	Understand the Software Development Life Cycle, algorithm and flow chart.	3	Understanding
Contents: Introduction to Computer Systems Introduction to computers, Evolution of computers, Characteristics of computers, Classification of computers, Generations; Basic Computer Architecture; Components of Computer Systems (I/O Devices), Computer Memory and its types; Data Representation basics; Computer Software, Relation Between Hardware and Software; Types of Software - System Software, Application Software; Free and open source software, Software Development Life Cycle; Introduction to Algorithm and Flow chart.			
CO2	Identify the different operating systems and their functions		
M2.01	Explain the different functions, types of operating systems, and the techniques used.	8	Understanding
M2.02	Compare the different operating systems.	2	Applying
	Series Test – I	1	

Contents:**Operating Systems**

Introduction to Operating Systems, Definition of OS, Layers of OS; Types of Operating Systems - single user & multi user OS, batch processing OS, Time sharing OS, Real Time OS; Operating System Techniques - Multiprogramming, Multitasking, Multithreading, Multiprocessing; Some Popular Operating Systems - DOS (Disk Operating System), UNIX Operating System, Linux, Microsoft Windows, Microsoft Windows NT, Mobile Operating systems

CO3	Utilize the concepts of Data Processing and Data Communication		
M3.01	Explain data and file processing methods	3	Understanding
M3.02	Describe the file organization methods and file operations	3	Understanding
M3.03	Identify the importance of database management system	1	Understanding
M3.04	Recognize the data communication modes, transmission methods and types of communication media.	4	Applying

Contents:

Business Data Processing and Data Communication

Data Processing; File Management System, Types of processing files; File Organization; File operations; Database Management System, its purpose and uses.

Basic Elements of a Communication System; Modes of data communication; Transmission Basics - Analog and Digital; Types of Data Transmission Media; Modems.

CO4	Interpret the basics of Computer Networks and Internet		
M4.01	Explain the different types of computer networks	4	Understanding
M4.02	Identify the different network topologies	3	Applying
M4.03	Describe about the basics of Internet.	3	Understanding
	Series Test – II	1	

Contents:**Computer Networks**

Need and Definition for Computer Communication Networks; Advantages of Networks; Types of Network; Network Topologies; Internet Basics - ISP, web server, web browser, domain names, World Wide Web, E-mail, Search Engines.

Text / Reference:

T/R	Book Title/Author
R ₁	Rajaraman, Fundamentals of Computers , Prentice Hall of India, 2003
R ₂	P.K Sinha, Priti Sinha, Computer Fundamentals , Kalyani Publishers, 2nd Edition
R ₃	Shelly, Business Data Communication , Thomson Learning, Bombay, 2003
R ₄	White, Data Communication & Computer Network , Thomson Learning, Bombay, 2012
R ₅	B. Ram, Computer Fundamentals , New Age International, 2000

Online Resources:

Sl.No	Website Link
1	https://www.javatpoint.com/computer-fundamentals-tutorial
2	https://www.tutorialspoint.com/computer_fundamentals/index.htm
3	https://swayam.gov.in/nd2_cec19_cs06/preview
4	http://examradar.com/communication-networking-summary-1/